# Au+Au 3.85 GeV: $\Lambda$–$^4$He CF in four $m_T$ slices

Centrality 0-20%, KFParticle

CF: 20 MeV/c; display 0-0.3 GeV/c; normalization 0.30-0.40 GeV/c.

The fixed boundaries are specified in YAML and coincide with stored 0.05 GeV/$c^2$ edges. They are not moved to equalize counts. Only the configured $m_T$ range is included; tails outside it are excluded. Excluded signal pairs for $0 \leq k^* < 0.3$ GeV/$c$: below range SE=0, ME=0; above range SE=252, ME=8313.

The same slices are used for both methods, SE, ME and sidebands.

| Slice | $m_T$ [GeV/$c^2$] | SE (0–0.3) | ME (0–0.3) | SE norm | ME norm |
|---|---|---|---|---|---|
| 1 | 2.40–2.60 | 17451 | 621134 | 30089 | 1095897 |
| 2 | 2.60–2.80 | 38934 | 1375699 | 49147 | 1767781 |
| 3 | 2.80–3.00 | 14079 | 504952 | 15306 | 552257 |
| 4 | 3.00–3.20 | 2427 | 86159 | 2582 | 91897 |

**Method 5 is an approximation, not an $m_T$-resolved mass fit.** The available histograms have no joint $(k^*, m_T, M_\Lambda)$ distribution. The same $m_T$-integrated, centrality-0-20% $P_{SE}(k^*)$ and $P_{ME}(k^*)$ are applied to all four slices. Purity grid: 50 MeV/$c$; five-bin sliding window: 250 MeV/$c$ in the interior. Each original 5 MeV/$c$ count is weighted before summing to 20 MeV/$c$.

Constant purity: $P = 0.96721065$ (the same inclusive proxy as before), with $C_{corr} = [C_{sig} - (1-P)C_{SB}]/P$. Sidebands and normalization are evaluated separately in each $m_T$ slice.

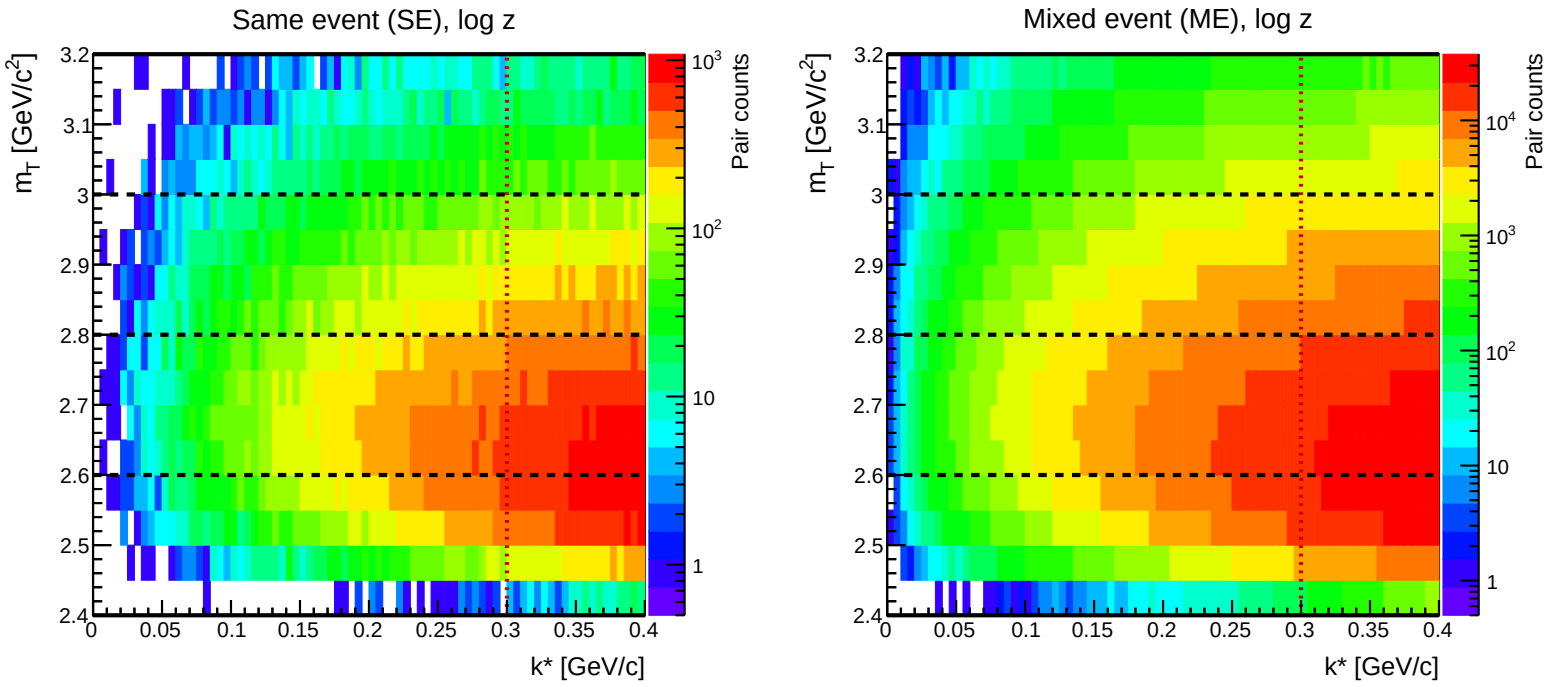
Approximate errors include shared purity reuse and normalization. Covariances between mass-fit backgrounds and observed counts, distinct overlapping fits, $m_T$ counts and inclusive purity, and reused event/mixed pairs are not available. Constant-$P$ errors are conditional on fixed $P$. Sparse/zero-SE bins do not have Poisson confidence intervals. The smallest SE signal bins contain only a few pairs despite the large slice totals. Exact candidate mass gates and whole mass-bin fit integrals differ slightly.

2021 P24iy only; 26,956/26,994 subjobs in this partial merge. The input ROOT and all production cuts are unchanged. Full provenance, statistics and caveats are in the adjacent README and JSON files. **Masked points (not zero):**

mT 3.00-3.20, constant_purity, k* 0.00-0.02. Zero ME denominators / invalid normalization are not filled.

$\Lambda$-$^4$He: $m_T$ split map | centrality 0-20%

Signal: ±3σ; original bins; horizontal lines: slice boundaries; normalization 0.30-0.40 GeV/c



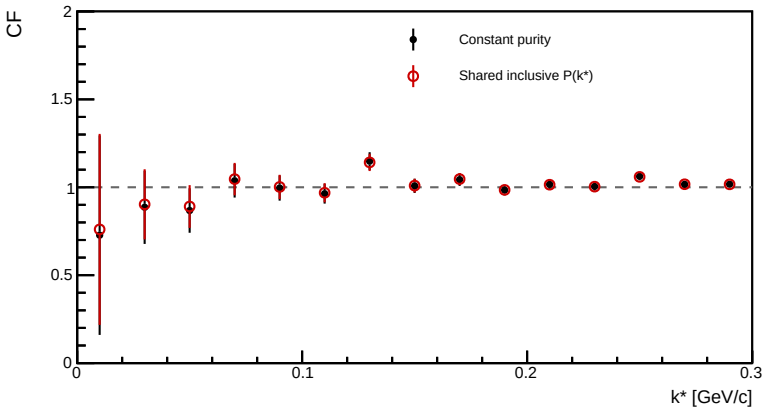

Black horizontal lines: supplied $m_T$ edges. Red vertical line: normalization start. No cut or reweighting is applied by this renderer.

# $\Lambda$-$^4$He: $m_T$-selected correlation functions | centrality 0-20%

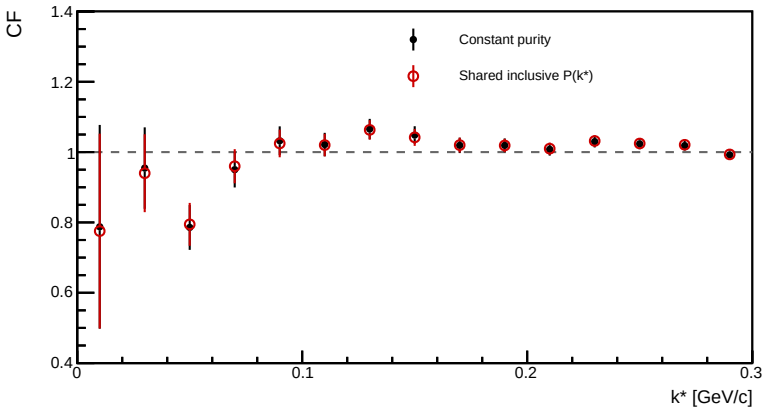
Signal: ±3σ; normalization 0.30-0.40 GeV/c; no hidden clipping of finite errors

Shared inclusive P(k*) (Method 5 approximation): purity is NOT fitted separately in each $m_T$ slice

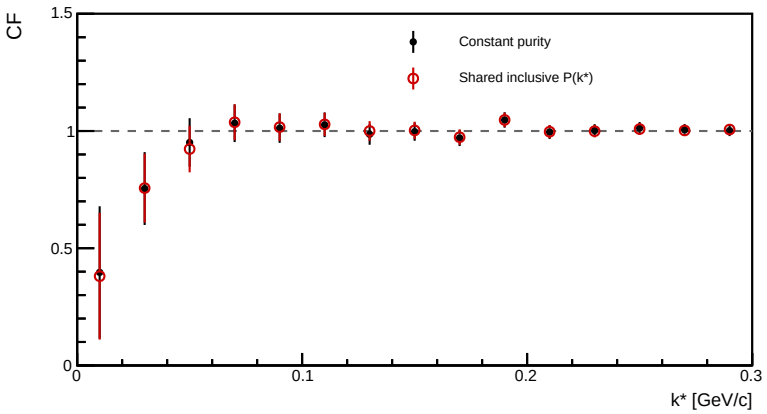
$m_T$ = 2.40-2.60 GeV/c$^2$



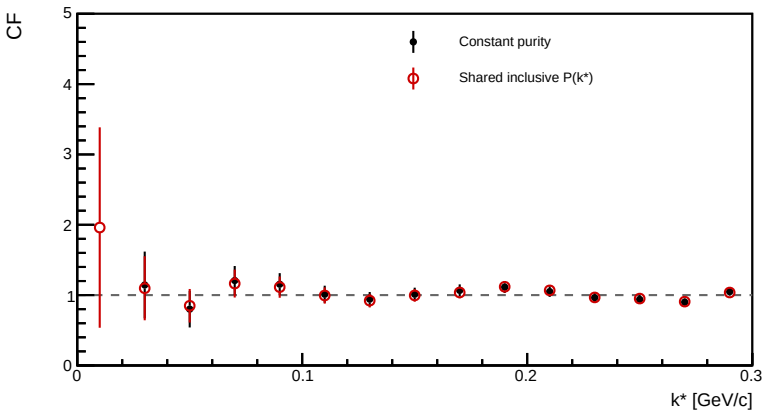

$m_T$ = 2.60-2.80 GeV/c$^2$




$m_T$ = 2.80-3.00 GeV/c$^2$




$m_T$ = 3.00-3.20 GeV/c$^2$



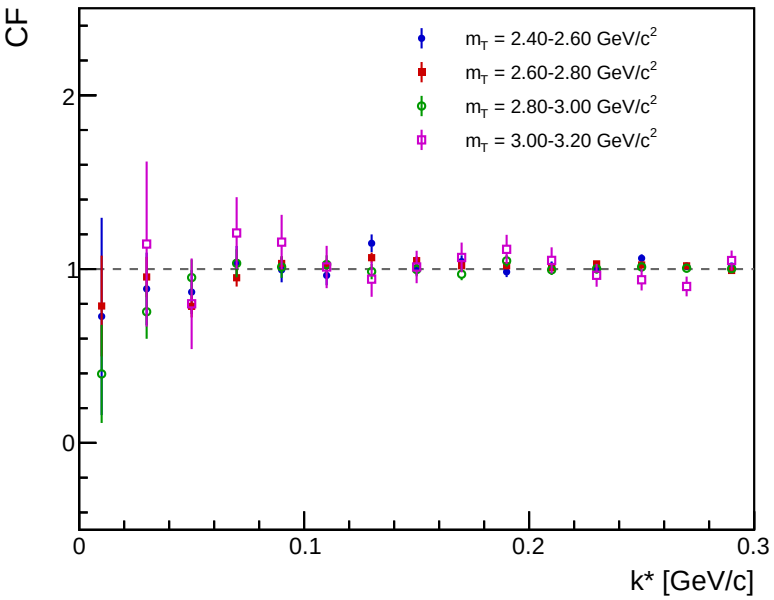

Invalid / missing bins are omitted, not set to zero. Normalization is shaded where visible.

$\Lambda$-$^4$He: $m_T$-selected correlation functions | centrality 0-20%

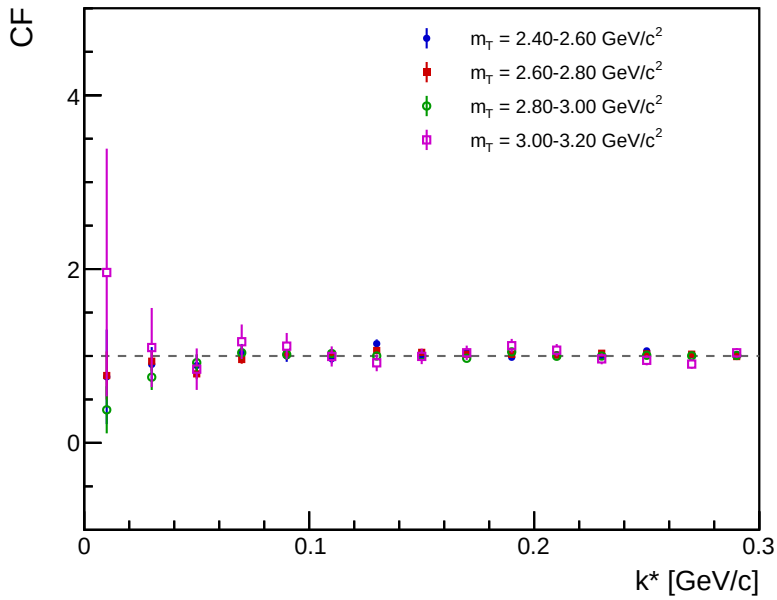
Signal: ±3σ; normalization 0.30-0.40 GeV/c; no hidden clipping of finite errors

Shared inclusive P(k*) (Method 5 approximation): purity is NOT fitted separately in each $m_T$ slice

Constant purity




Shared inclusive P(k*)
(Method 5 approximation)




The same supplied CF values are reused; this renderer performs no fit or background correction.